

Name _____ Surname _____ Date ____/____/____

Midterm test, Term 1 — Speed, energy and work

Version C

Time: 75 minutes. Calculator and formula sheet (last page) allowed. Use $g = 10 \text{ N/kg}$. Write on separate sheets, show every step of your working and write the unit in each answer. Give answers to 3 significant figures where needed. Total: 55 marks.

 **Lifebuoy.** The parts marked with the lifebuoy are worth 33 marks out of 55, that is 60%: if you have completed all of them, you are safe. The other parts take you beyond the pass mark.

Exercise 1 — True or false

 Write T or F. No explanation needed. Each statement is worth 1 mark, 4 marks in total.

- (a) When an object falls, gravitational potential energy is destroyed.
- (b) Doubling the height of an object of the same mass doubles its gravitational potential energy.
- (c) To change a speed from km/h to m/s you multiply by 3.6.
- (d) A force that moves an object through a distance in the direction of the force does work on it.

Exercise 2 — Multiple choice

 Only one answer is correct. Each question is worth 1 mark, 8 marks in total.

1. The energy stored in petrol is:
A nuclear energy B elastic (strain) energy C kinetic energy D chemical energy
2. A runner covers 400 m in 50 s. Her average speed is:
A 0.125 m/s B 8.0 m/s C 20 000 m/s D 350 m/s
3. A straight sloping line on a distance–time graph shows that the object moves:
A at constant speed B at rest C with increasing speed D with decreasing speed
4. A box slides down a rough ramp. Which statement is correct?
A All the gravitational potential energy becomes kinetic energy. B Friction destroys energy.
C Some of the gravitational potential energy is transferred to thermal energy. D The box gains energy from the ramp.
5. The speed of a cart is halved. Its kinetic energy becomes:
A one half B double C the same D one quarter
6. Which situation does NOT involve work done on the object?
A a crane lifts a container B a shelf supports a pile of books C a woman pulls a sledge
D a boy pushes a car that moves
7. A force of 30 N moves an object 0.20 m in the direction of the force. The work done is:
A 150 J B 30.2 J C 6.0 J D 0.0067 J
8. A 2.0 kg book is lifted 3.0 m. Its gravitational potential energy increases by:
A 60 J B 6.0 J C 0.67 J D 5.0 J

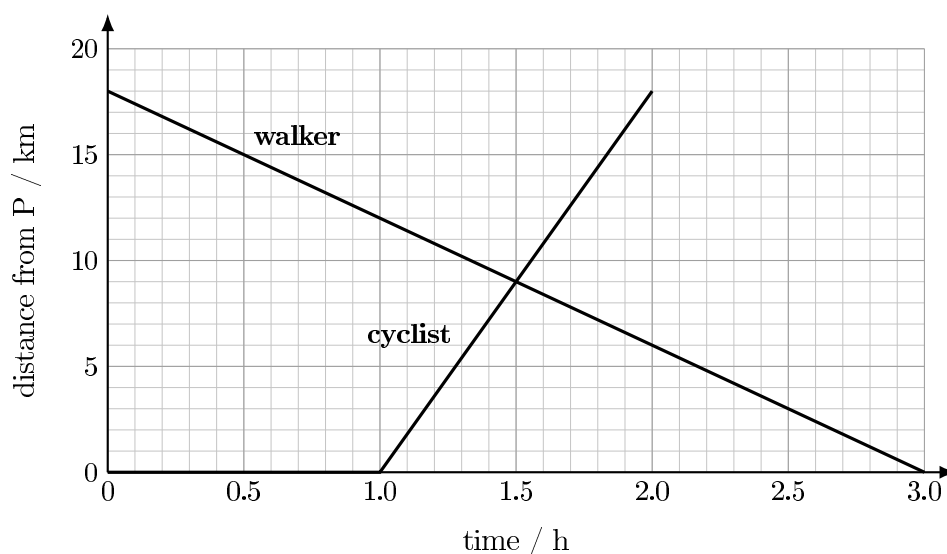
Exercise 3 — Energy stores

⚽ Complete the table. In each event, name the energy store that decreases and the one that increases. 1 mark for each row, 3 marks in total.

Event	Store that decreases	Store that increases
A ball is thrown upwards, while it rises.		
A compressed spring in a toy dart gun is released.		
A diver climbs the ladder to the top board.		

Exercise 4 — Two travellers

A walker and a cyclist travel on the same straight path between village P and village Q. The graph shows the distance of each of them from P. The walker starts from Q and the cyclist starts from P at the same moment.



- ⚽ Read the graph: for how long does the cyclist stay at rest at the start, and at what time and at what distance from P do the two meet? [2]
- Calculate the speed of the cyclist while she is moving, in km/h. [2]
- Calculate the average speed of the cyclist for the whole journey, including the rest, in km/h. [2]
- Calculate the speed of the walker in m/s. [2]

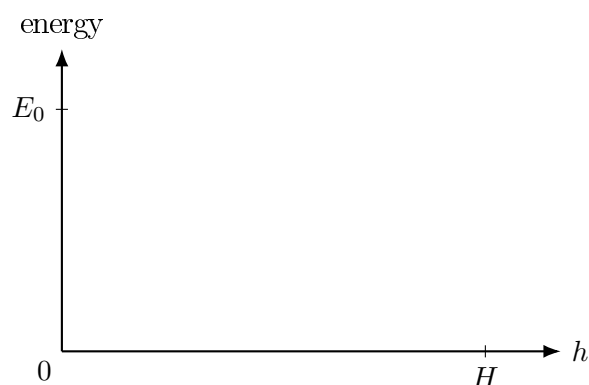
Exercise 5 — A motorbike brakes

A motorbike and its rider have a total mass of 250 kg and travel at 20 m/s. The rider brakes and they stop after 40 m. Assume that the braking force is constant.

- (a)  In which energy store does the kinetic energy mainly end up? [1]
- (b) Calculate the kinetic energy before braking. [2]
- (c) Calculate the braking force. [2]
- (d)  The brakes are now changed so that the braking force is twice as big. The mass and the speed are the same. The stopping distance becomes: the same, half as long, one quarter as long, twice as long. Circle the correct answer and give a reason in one line. [1]

Exercise 6 — A tennis ball thrown upwards

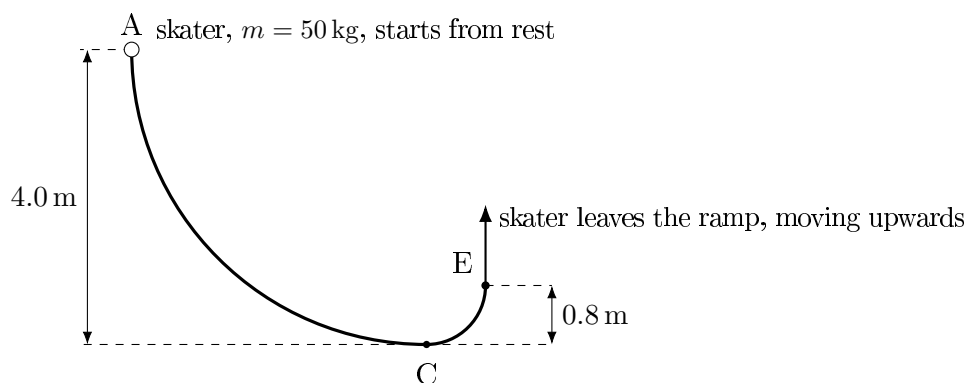
A tennis ball of mass 0.060 kg is thrown vertically upwards from the ground at 20 m/s. Ignore air resistance. On the axes below, E_0 is the kinetic energy at launch and H is the maximum height.



- (a)  Draw and label how the kinetic energy and the gravitational potential energy change with the height h , from $h = 0$ to $h = H$. [2]
- (b)  Write, in symbols and without numbers, the equation that gives the maximum height H . [2]
- (c) Calculate the kinetic energy at launch. [2]
- (d) Calculate the maximum height. [2]
- (e)  The same object is thrown with twice the speed. The maximum height becomes: the same, twice as big, four times as big, eight times as big. Circle the correct answer and give a reason in one line. [2]

Exercise 7 — The half-pipe

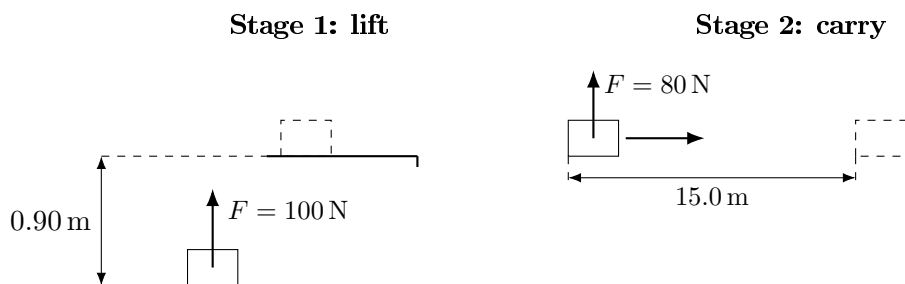
A skater of mass 50 kg starts from rest at A, at the top of a half-pipe, 4.0 m above the lowest point C. She rolls down, passes C and goes up the right-hand side, leaving the ramp at E, 0.8 m above C, moving vertically upwards. Ignore friction and air resistance.



- Without calculating, say how high above C the skater rises before she starts to fall, and justify your answer in one line. [2]
- Write, in symbols and without numbers, the equation that gives her speed v at E in terms of the heights h_A and h_E above C. [2]
- Calculate the gravitational potential energy lost between A and E. [2]
- Calculate her speed at E. [2]

Exercise 8 — Lifting and carrying a box

A student lifts a box of mass 8.0 kg from the floor to a table 0.90 m above it (stage 1). Her hands push upwards on the box with a constant force of 100 N, and the box is still moving upwards when it reaches the height of a table. In stage 2 she walks 15.0 m across the school hall, holding the box at that height with a force of 80 N from her hands, pointing upwards.



- Write, in symbols, the equation for the work done by her hands and the equation for the gain in gravitational potential energy of the box in stage 1. [2]
- Calculate the work done by her hands on the box and the gain in gravitational potential energy of the box in stage 1. [2]
- The work done is greater than the gain in gravitational potential energy. In which energy store does the difference end up? [1]
- State the work done by her hands on the box in stage 2. [1]
- Explain your answer to (d) in one or two lines. (Her muscles use chemical energy even while she carries the box: here we ask only about the work done on the box.) [2]